



Calculating theoretical probabilities from Venn diagrams (one event)

Task A: Theoretical probabilities from Venn diagrams

1) Use the Venn diagram to write the probabilities.

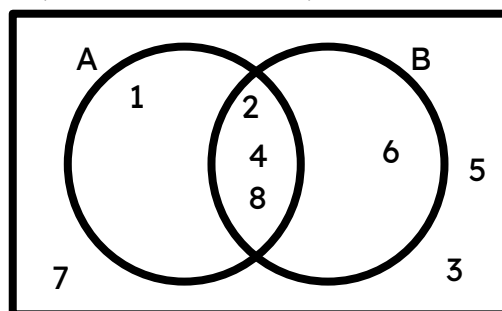
$$a) P(A) = \frac{4}{8} = \frac{1}{2} = 50\% = 0.5$$

$$b) P(B) = \frac{4}{8} = \frac{1}{2} = 50\% = 0.5$$

$$c) P(\text{not } B) = \frac{4}{8} = \frac{1}{2} = 50\% = 0.5$$

$$d) P(\text{only } A) = \frac{1}{8} = 12.5\% = 0.125$$

$$\xi = \{1, 2, 3, 4, 5, 6, 7, 8\}$$



2) Use the Venn diagram to write the probabilities.

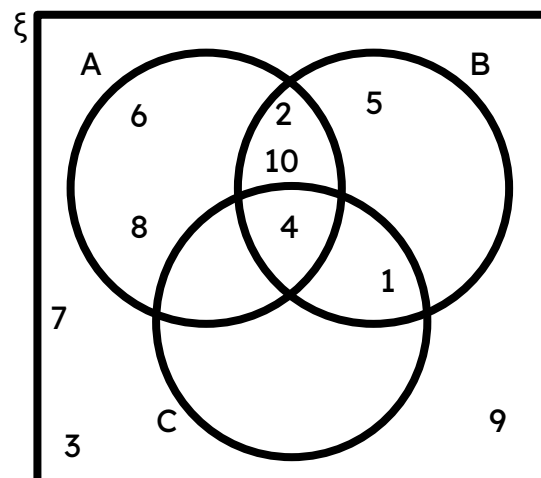
$$a) P(A) = \frac{5}{10} = \frac{1}{2} = 50\% = 0.5$$

$$b) P(B) = \frac{5}{10} = \frac{1}{2} = 50\% = 0.5$$

$$c) P(C) = \frac{2}{10} = \frac{1}{5} = 20\% = 0.2$$

$$d) P(\text{only } B) = \frac{1}{10} = 10\% = 0.1$$

$$e) P(\text{not } C) = \frac{8}{10} = \frac{4}{5} = 80\% = 0.8$$



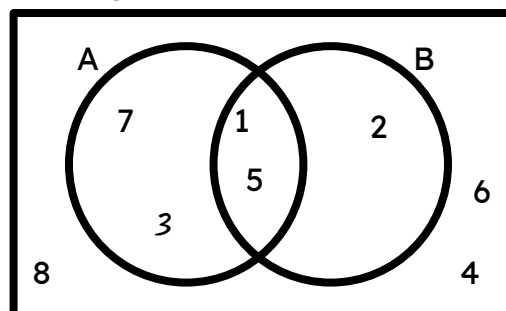
3)

a) Complete the Venn diagram given these probabilities.

$$P(A) = \frac{1}{2}$$

$$P(B) = \frac{3}{8}$$

$$\xi = \{\text{integers 1 to 8 inclusive}\}$$



b) Work out $P(\text{only } A)$

$$\frac{2}{8} = \frac{1}{4} = 25\% = 0.25$$

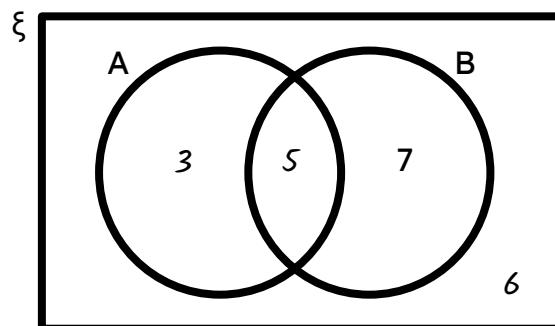
4) This Venn diagram shows the number of outcomes in each event.

There are 21 outcomes in the sample space.

Complete the Venn diagram given these probabilities.

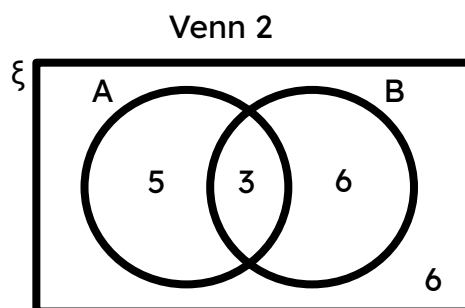
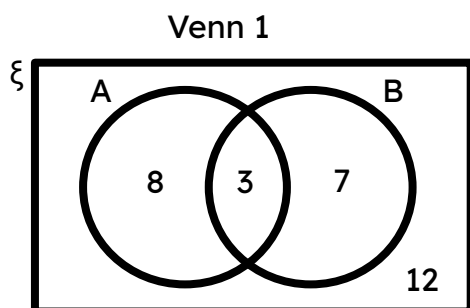
$$P(\text{not } A) = \frac{13}{21}$$

$$P(\text{only } A) = \frac{1}{7}$$



Task B: Comparing probabilities

1) Compare the probabilities from each Venn diagram. The Venn diagrams show the number of outcomes.



a) $P(A)$

Venn 1:

$$P(A) = \frac{11}{30} = \frac{22}{60}$$

Venn 2:

$$P(A) = \frac{8}{20} = \frac{24}{60}$$

So event A is more likely in Venn 2.

b) $P(\text{only } A)$

Venn 1:

$$P(\text{only } A) = \frac{8}{30} = \frac{16}{60}$$

Venn 2:

$$P(\text{only } A) = \frac{5}{20} = \frac{15}{60}$$

So only event A is more likely in Venn 1.

c) $P(\text{not } B)$

Venn 1:

$$P(\text{not } B) = \frac{20}{30} = \frac{40}{60}$$

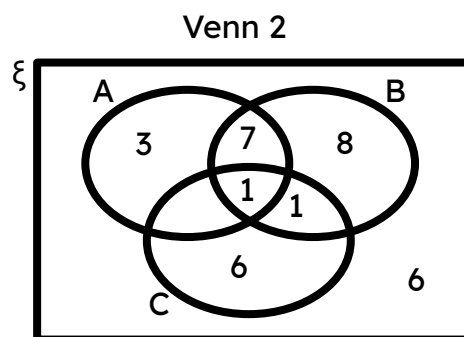
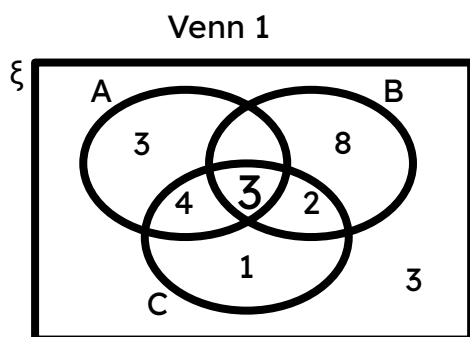
Venn 2:

$$P(\text{not } B) = \frac{11}{20} = \frac{33}{60}$$

So not event B is more likely in Venn 1.



2) Compare the probabilities from each Venn diagram. The Venn diagrams show the number of outcomes.



a) $P(A)$

Venn 1:

$$P(A) = \frac{10}{24} = \frac{40}{96}$$

Venn 2:

$$P(A) = \frac{11}{32} = \frac{33}{96}$$

So event A is more likely in Venn 1.

b) $P(C)$

Venn 1:

$$P(C) = \frac{10}{24} = \frac{40}{96}$$

Venn 2:

$$P(C) = \frac{8}{32} = \frac{24}{96}$$

So event C is more likely in Venn 1.

c) $P(\text{not } B)$

Venn 1:

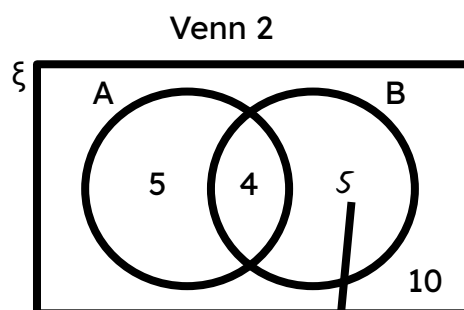
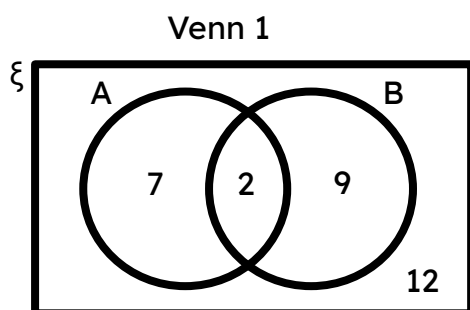
$$P(\text{not } B) = \frac{11}{24} = \frac{44}{96}$$

Venn 2:

$$P(\text{not } B) = \frac{15}{32} = \frac{45}{96}$$

So not event B is more likely in Venn 2.

3) Complete the second Venn diagram with number of outcomes such that you are more likely to select an outcome from event B in Venn 2.



Any integer greater than 4

